Probability and Random Processes

Branching Processes.
Exercises

Branching processes

1. Consider a branching process that begins with a single ancestor. The offspring distribution has the following probability function:

$$\mathbb{P}(X=0)=\frac{1}{4},\quad \mathbb{P}(X=1)=\frac{1}{2},\quad \mathbb{P}(X=2)=\frac{1}{8},\quad \mathbb{P}(X=3)=\frac{1}{8}.$$

What is the probability

(a) that the process becomes extinct in generation 2 (that is, the ancestor has no grandchildren)?

(b) of eventual extinction?

2. At each unit of time, a cell either dies with probability p or subdivides with probability $1-p$. Let X_n be the number of cells at time n.

(a) If $X_0=1$, what is the maximum value of p for which the probability of extinction is at most $1/4$?

(b) If $X_0=k$ and $p=1/4$, what is the minimum value of k such that the probability of extinction is at most $1/4$?

3. Consider a branching process that starts with a single ancestor. The probability generating function of the offspring distribution is

$$G(s)=\frac{1}{2-s^2}.$$

(a) What is the probability that the first generation has 4 individuals?

(b) Compute the probability of extinction.

(c) What is the probability that the process is already extinct by the third generation?

4. *(A. Gut's book)* Consider a branching process with one ancestor. Suppose that the generating function of the offspring distribution is

$$G(s)=\frac{p^2}{(1-qs)^2},$$

where $0<p=1-q<1$. What is the probability

(a) of extinction?

(b) that the process is extinct in generation number 2 (i.e., the ancestor does not have any grandchildren)?

5. *(A. Gut's book)* The following model can be used to describe the number of women (mothers and daughters) in a given area. The number of mothers is a random variable $X\sim \mathsf{Po}(\lambda)$. Independently of the others, every mother gives birth to a $\mathsf{Po}(\mu)$-distributed number of daughters. Let Y be the total number of daughters and hence $Z=X+Y$ be the total number of women in the area.

(a) Find the generating function of Z.

(b) Compute $\mathbb{E}(Z)$ and $\mathrm{Var}(Z)$.

6. At each time unit, a virus replicates X copies of itself, where $\mathbb{P}(X = k) = p^k$ for $k \geqslant 1$ and $0 < p \leqslant 1/2$. (Thus, the probability of no replication is $1 - p/q$, where $q = 1 - p$.) Assume that at time 0, there is a single originating virus.

(a) Prove that the probability generating function of X is given by $G_X(s) = \dfrac{1}{1 - ps} - \dfrac{p}{q}$.

(b) Show that the probability of extinction is $1/p - 1/q$ if $p > (3 - \sqrt{5})/2$ and 1 otherwise.

(c) If $p = 1/4$, what is the expected total number of viruses at the end?

7. A politician convinces X collaborators to vote positively in a poll. In each round, each persuaded collaborator independently convinces X additional collaborators, where X is a random variable such that $\mathbb{P}(X = 0) = \mathbb{P}(X = 2) = 1/4$ and $\mathbb{P}(X = 1) = 1/2$.

(a) What is the probability that there will be no more collaborators voting for the politician after a certain round?

Let Z_n be the number of votes raised by the politician in round n, and let T_n be the total number of votes raised over rounds $0, 1, \ldots, n$. Assume $Z_0 = T_0 = 1$, meaning the politician votes for himself.

(b) Find $\mathbb{E}(Z_n)$. What is the expected number $\mathbb{E}(T_n)$ of positive votes raised by round n?

(c) Show that the probability generating function of T_n satisfies the equation

$$G_{T_n}(s) = s\, G_X\big(G_{T_{n-1}}(s)\big)\,.$$

[Hint: Condition on the number of votes raised in the first round to find $\mathbb{E}\left(s^{T_n} \mid Z_1 = k\right)$.]

(d) Compute the generating function of T_2. What are the probabilities of the most and least probable values of this variable? Find the variance of T_2. What is the probability that T_2 exceeds its expected value?

8. In each generation of a branching process, each individual produces X copies of itself and dies. The number X is a random variable that follows a Poisson distribution with parameter λ. We start with a single individual at time 0.

(a) Discuss in terms of λ the behaviour of the probability of ultimate extinction.

(b) Let $\lambda = 2$. Find the probability of ultimate extinction. If there were k originating individuals at time 0, what is the minimum value of k for which the probability of extinction is less than 1%? [*Hint:* A solution of the equation $xe^{-x} = 2e^{-2}$ is $x \approx 0.4064$.]

(c) If $\lambda = 2$ and Z_2 is the number of individuals in the second generation, find $\mathbb{P}(Z_2 = 0)$ and $\mathbb{P}(Z_2 = 1)$.

9. In a synchronous process, the number of active tasks at a given time period k, where $k = 0, 1, 2, \ldots$, is represented by the random variable N_k, with the initial condition $N_0 = 1$. Each task remains active for exactly one time period. During period k, each active task generates (independently of the others) a random number of new tasks that follow a binomial distribution, denoted as $X \sim \mathsf{Bin}(n, p)$. These new tasks will be active in period $k + 1$.

(a) Let $n = 3$ and $p = 1/2$. Calculate the probability of at least an active task in each period k, $k \geqslant 0$. Calculate also $\mathbb{P}(N_2 = 0)$ and $\mathbb{P}(N_2 = 1)$.

(b) For general values of n and p, determine $\mathbb{E}(N_2)$ and prove that $\mathrm{Var}(N_2) = npq(n^2p^2 + np)$, where $q = 1 - p$.

10. *(A. Gut's book)* Suppose that the offspring distribution in a branching process is a geometric distribution such that $\mathbb{P}(X = k) = q^k p$, $k = 0, 1, 2, \ldots$, and let $X(n)$ be the number of individuals in generation n, $n = 0, 1, 2, \ldots$

(a) What is the probability of extinction?

Now suppose that $p = 1/2$ and set $G_n(s) = G_{X(n)}(s)$.

(b) Show that

$$G_n(s) = \frac{n - (n-1)s}{(n+1) - ns}, \quad n = 1, 2, \ldots$$

(c) Show that

$$\mathbb{P}(X(n) = k) = \begin{cases} \dfrac{n}{n+1} & \text{for } k = 0 \\ \dfrac{n^{k-1}}{(n+1)^{k+1}} & \text{for } k = 1, 2, \ldots \end{cases}$$

(d) Calculate $\mathbb{P}(X(n) = k \mid X(n) > 0)$.

Finally, suppose that the population becomes extinct at generation number N.

(e) Show that

$$\mathbb{P}(N = n) = G_{n-1}\left(\frac{1}{2}\right) - G_{n-1}(0), \quad n = 1, 2 \ldots$$

(f) Show that $\mathbb{P}(N = n) = 1/(n(n+1))$, $n = 1, 2, \ldots$ (and hence that $P(N < \infty) = 1$, i.e., the probability of ultimate extinction is 1).

(g) Compute $\mathbb{E}(N)$.

Answers

1. (a) $197/512 \approx 0.385$; (b) $-1 + \sqrt{3} \approx 0.732$.
2. (a) $p \leqslant 1/5$; (b) $k \geqslant 2$.
3. (a) $1/8$; (b) $(-1 + \sqrt{5})/2 \approx 0.618$; (c) $49/82 \approx 0.598$.
4. (a) If $0 < p < 2/3$, then $d = \frac{1}{q} - \frac{1}{2} - \sqrt{\frac{1}{q} - \frac{3}{4}}$ and if $p \geqslant 2/3$, then $d = 1$; (b) $d_2 = p^2/(1 - p^2 q)^2$.
5. (a) $G_Z(s) = \exp\left(\lambda\left(se^{\mu(s-1)} - 1\right)\right)$; (b) $\mathbb{E}(Z) = \lambda(1 + \mu)$; $\mathrm{Var}(Z) = \lambda(1 + 3\mu + \mu^2)$.
6. (c) $9/5$.
7. 1; $\mathbb{E}(Z_n) = 1$; $\mathbb{E}(T_n) = n + 1$; $G_{T_2}(s) = \frac{1}{64}\left(16s + 8s^2 + 17s^3 + 12s^4 + 6s^5 + 4s^6 + s^7\right)$; $17/64$ and $1/64$; $\mathrm{Var}(T_2) = 5/2$; $23/64$.
8. (a) if $\lambda \leqslant 1$, then $d = 1$, and if $\lambda > 1$, then $d < 1$; (b) 0.2032; (c) $\mathbb{P}(Z_2 = 0) \approx 0.177$, $\mathbb{P}(Z_2 = 1) \approx 0.0960$.
9. (a) $3 - \sqrt{5} \approx 0.764$; $\mathbb{P}(N_2 = 0) = \mathbb{P}(N = 1) = 729/4096 \approx 0.178$; (b) $\mathbb{E}(N_2) = n^2 p^2$.
10. (a) If $p < q$ then $d = p/q < 1$ and if $p \geqslant q$, then $d = 1$; (d) $(1/(n+1))(n/(n+1))^{k-1}$, $k \geqslant 1$; (g) $\mathbb{E}(N) = \infty$.